Chapter-7

Application of Group theory to molecular vibration.

⇒ Application of the group theory and molecular symmetry:

(1) construction of hybrid orbitals.

(2) Finding orbitals suitable for π-bonding

(3) Determine the symmetry of LCAO molecular orbitals.

(4) Construction of qualitative molecular orbital correlation diagrams.

(5) Find the symmetry species of the normal modes of vibration of the molecule of a given symmetry.

(6) Find the no. of infrared and Raman active vibrations in a molecules.

(7) Find the no. of band in the IR and the Raman active molecules.

⇒ The use of group theory can be summarised in the following three rules:

(1) Use an appropriate basis to generate a reducible representation of the point group.

(2) Reduce the representation to its constituent irreducible representations.

(3) Interpret the result.

⇒ If there are n atoms in a molecule. There are 3N possible movements of its atoms.

- translations, (along the three co-ordinate axes). total movement : 3
- Rotations,
  - → total movement 3 (2 for the linear molecule). Rotation about the axes.
- The remaining (3N-6) or 3N-5 (for linear molecule) for vibration.

⇒ How many there of vibration in XeF_4?

There are 5 atoms, N = 5

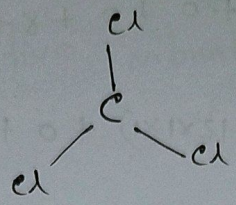
Total movement of the atoms in a molecule, = 3N
= 15

There is not a linear molecule. So, 3 translation, 3 rotation are possible.

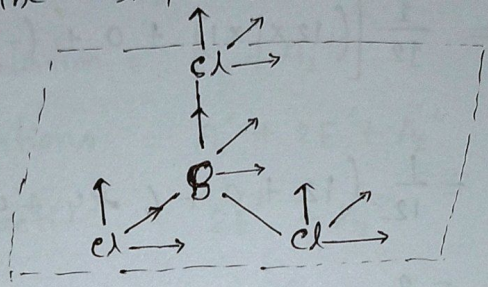
So remaining, = 3N-6
= 15-6
= 9 are for vibrations.

(1) Find the number and symmetry species of the Raman and IR active vibration of boron trichloride (D_{3h}):

Structure:



Here, N = 4, The basis of the representation is the set of the 3×4 = 12 arrows.



The point group of the BCl_3 is D_{3h}. The complete set of ~~the~~ character of the reducible representation is:

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ	12	0	−2	4	−2	2

According to the character table of D_{3h} point group:

No. of A_1' = $\frac{1}{12}[(12\times1\times1) + 0 + (-2\times3\times1) + (4\times1\times1) + (-2\times2\times1) + (2\times3\times1)]$

= $\frac{1}{12}(12-6+4-4+6) = 1$

No. of A_2' = $\frac{1}{12}[(12\times1\times1) + 0 + (-2\times3\times(-1)) + (4\times1\times1) + (-2\times2\times1) + (2\times3\times(-1))]$

= $\frac{1}{12}(12+6+4-4+6) = 1$

No. of $E' = \frac{1}{12}[(12\times1\times2) + 0 + (-2\times0\times3) + (4\times2\times1) + (-2\times2\times-1) + (2\times0\times3)]$

$= \frac{1}{12}(24 + 0 + 0 + 8 + 4 + 0) = 3$

No. of $A'' = \frac{1}{12}[(12\times1\times1) + 0 + (-2\times3\times1) + (4\times1\times-1) + (-2\times2\times-1) + (2\times3\times-1)]$

$= \frac{1}{12}(12 - 6 - 4 + 4 - 6) = 0$

No. of $A_2'' = \frac{1}{12}[(12\times1\times1) + 0 + (-2\times3\times(-1)) + (4\times1\times(-1)) + (-2\times2\times(-1)) + (2\times3\times2)]$

$= \frac{1}{12}(12 + 0 + 6 - 4 + 4 + 6)$

$= 2$

No. of $E'' = \frac{1}{12}[(12\times1\times2) + 0 (-2\times3\times0) + (4\times(-2)\times1) + (-2\times2\times1) + (2\times3\times0)]$

$= \frac{1}{12}(24 + 0 + 0 - 8 - 4 + 0) = 1$

So, $\Gamma = A_1' + A_2' + 3E' + 2A_2'' + E''$

There are total molecular movement = 12

[R_x, R_y, R_z belongs to the A_2, E'' representation] [x, y, z axis belongs to the E', A_2'' representation]

According to the character table of C_{3h}.

Total movement: $A_1' + A_2' + 3E' + 2A_2'' + E''$

Rotations: $A_2 + E''$

Translations: $E' + A_2''$

$\therefore$ vibration $= A_1' + 2E' + A_2''$

There are total 6 mode of vibration is possible,

The IR active vibrations are = $2E' + A_2''$

The Ramman active vibrations are = $A_1' + 2E'$

Answer:

$\Gamma = A_1' + A_2' + 3E' + 2A_2'' + E''$

Rotations = $A_2' + E''$

Translation = $E' + A_2''$

vibrations = $A_1' + 2E' + A_2''$

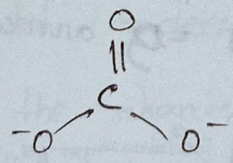
IR active = $2E' + A_2''$

Raman active = $A_1' + 2E'$

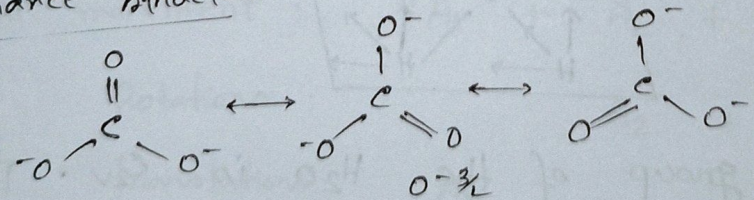
Tutorial – 2020

Find the no. of symmetry species of the Raman and infrared active vibration of CO_3^{2-} ion.

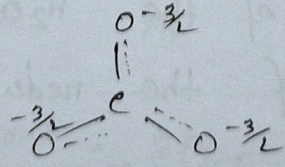
Structure:



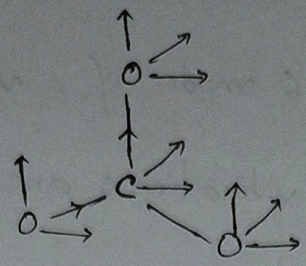
Resonance structure:

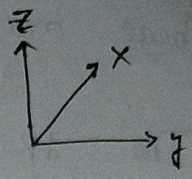


overall



Here, N = 4, the basis of the representation is the set of the 3×4 = 12 arrows.





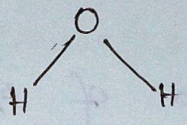
The point group of the CO_3^{2-} is D_{3h}. The complete set of the chanacter of the reducible nepnesentation is,

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ	12	0	-2	4	-2	2

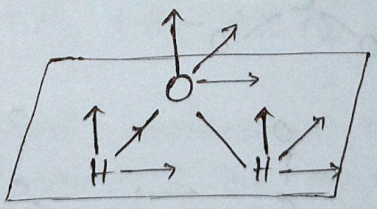
$\Rightarrow$ Same as BCl_3.

$\Rightarrow$ Find the number and symmetny speciens of the Raman and infnaned active vibnation of H_2O molecule.

Structune:



Here, N=3, The basin of the nepnesentation is the set of the $3\times3 = 9$ anrows.



The point group of the H_2O is C_{2v}. The complete set of chanacten of the neducible nepnesentation is,

C_{2v}	E	C_2	σ	σ'
Γ	9	-1	3	1

Aeconding to the chanacten table of C_{2v} point group

No. of $A_1 = \frac{1}{4}[(9\times1\times1) + (-1\times1\times1) + (3\times1\times1) + (1\times1\times1)]$

$= \frac{1}{4}(9-1+3+1) = 3$

No. of $A_2 = \frac{1}{4}[(9\times1\times1) + (-1\times1\times1) + (3\times1\times-1) + (1\times1\times-1)]$

$= \frac{1}{4}(9-1-3-1) = 1$

No. of $B_1 = \frac{1}{4}[(9\times1\times1) + (-1\times1\times-1) + (3\times1\times1) + (1\times1\times-1)]$

$= \frac{1}{4}(9+1+3-1) = 3$

No. of $B_2 = \frac{1}{4}[(9\times1\times1) + (-1\times-1\times1) + (3\times-1\times1) + (1\times1\times1)]$

$= \frac{1}{4}[(9+1-3+1]$

$= 2$

So, $\Gamma = 3A_1 + A_2 + 3B_1 + 2B_2$

There are total molecular movement = 9

According to the character table of C_{2v},

[R_x, R_y, R_y belongs to A_2, B_1, B_2 representation] [x, y, z belongs to A_1, B_2, B_1 representation]

Total movement :	$3A_1$	+ A_2	+ $3B_1$	+ $2B_2$	
Rotations :		A_2	+ B_1	+ B_2	
translations :	A_1		B_1	B_2	

$\therefore$ vibrations : $2A_1 + B_1$

There are total 3 modes of vibration in possible.

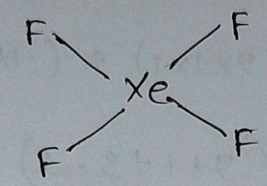
The IR active vibrations are = $2A_1 + B_1$

The Raman active vibrations are = $2A_1 + B_1$

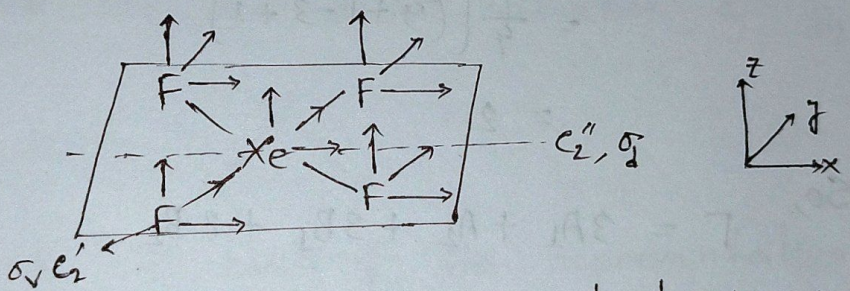
In, H_2O: There 3-IR and 3-Raman coincidences. The frequency of the IR absorption and the raman shift are identical.

⇒ Find the no. of symmetry species of the Raman and infrared active vibration of XeF_4 molecule.

Structure:



Here, N=5, the basis of the representation is the set of the $3\times5 = 15$ arrows.



The point group of the XeF_4 molecule is D_{4h}. The complete set of character of the reducible representation is,

D_{4h}	E	$2C_4$	$(C_2 = C_4^2)$	$2C_2'$	$2C_2''$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$
	15	1	-1	-3	-1	-3	-1	5	3	1

According to the character table of D_{4h} point group.

No. of $A_{1g} = \frac{1}{16}\left[(15\times1\times1) + (1\times2\times1) + (-1\times1\times1) + (-3\times2\times1) + (-1\times2\times1) + (-3\times1\times1) + (1\times2\times1) + (5\times1\times1) + (3\times2\times1) + (1\times2\times1)\right]$

$= \frac{1}{16}(15+2-1-6-2-3-2+5+6+2$

$= \frac{16}{16} = 1$

So, $\Gamma = A_{1g} + A_{2g} + B_{1g} + B_{2g} + E_g + 2A_{2u} + B_{2u} + 3E_u$

There are the total molecular movement = 15

According to the character table of C_{2h},

Rotation belongs to $A_{2g} + E_g$ (R_x, R_y, R_z) in the character table.

Total movement = $A_{1g} + A_{2g} + B_{1g} + B_{2g} + E_g + 2A_{2u} + B_{2u} + 3E_u$

Rotations = $A_{2g} + E_g$

Translations = $A_{2u} + E_u$

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vibrations = $A_{1g} + B_{1g} + B_{2g} + A_{2u} + B_{2u} + 2E_u$

There are total 9 mode of vibration is possible.

The IR active vibration are = $A_{2u} + 2E_u$

Raman active vibrations = $A_{1g} + B_{1g} + B_{2g}$

The IR active bands is 3. and the naman active band is 3. This band is no coincidences to each other.

This is the example of general effect called the mutual exclusion principle. This is applicable, that contains center of symmetry.

⇒ "Breathing" modes in which all four fluonines move out together and then in together. This modes of vibration is fully symmetric and belongs to A_{1g} irreducible representation.

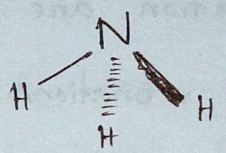
IR and Raman active rules:

1] A vibration will be infrared active if it belongs to the same symmetry species as a component of dipole moment. i.e. to the same species as either x, y or z.

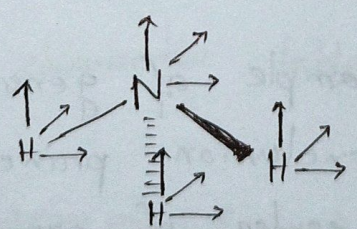
2] A vibration will be the Raman active if it belongs to the same symmetry species as a component of polarisability, i.e. to one of the binary products $x^2, y^2, z^2, xy, xz, yz$ or to combination of products such as $x^2 - y^2$.

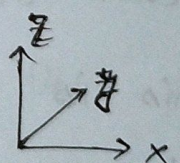
Find the no. of symmetry species of the Raman and infrared active vibration of NH_3 molecule.

Structure:



Here, $N = 4$, the basis of the representation is the set of the $3 \times 4 = 12$ arrows.





The point group of the BNH_3 is C_{3v}. The complete set of character of the reducible representation is,

C_{3v}	E	$2C_3$	$3\sigma_v$
Γ	12	0	2

→ C_3 এর মধ্যে 0

According to the character table of C_{3v} point group.

No. of $A_1 = \frac{1}{6}[(12\times1\times1) + 0 + (2\times3\times1)]$

$= \frac{1}{6}\times(12+6) = 3$

No. of $A_2 = \frac{1}{6}[(12\times1\times1) + 0 + (2\times3\times-1)]$

$= \frac{1}{6}(12+0-6) = 1$

No. of $E = \frac{1}{6}[(12\times2\times1) + 0 + 0]$

$= 4$

So, $\Gamma = 3A_1 + A_2 + 4E$

There are to total molecular movement $= 12$

According to the character table of C_{3v}

Total movement $= 3A_1 + A_2 + 4E$

Translation $= A_1 \quad E$

Rotations $= A_2 \quad E$

vibnations $= 2A_1 + 2E$

There are 6 mode of vibnations in possible.

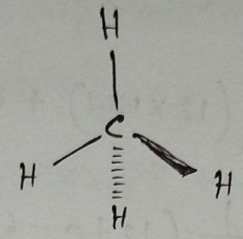
The IR active vibnations are $= 2A_1 + 2E$

The Raman active vibration are = $2A_1 + 2E$

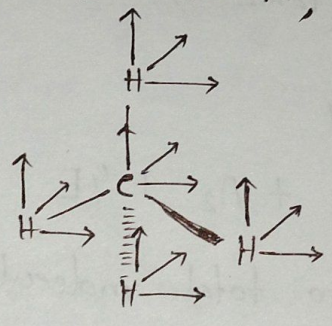
Therefore, there all 6 vibrations are IR and Raman active. IR band → 4
Raman band → 4

⇒ Find the number and symmetry species of the Raman and IR active vibration in methane.

Structure:



Here, N = 5, The basis of the representation is the set of the $3 \times 5 = 15$ arrows,



The point group of the CH_4 is T_d. The complete set of the character of the reducible representation is:

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$
Γ	15	0	-1	-1	3

According to the character table of T_d point group:

No. of $A_1 = \frac{1}{24}\left[(15 \times 1 \times 1) + 0 + (-1 \times 3 \times 1) + (-1 \times 6 \times 1) + (3 \times 6 \times 1)\right]$

$= \frac{1}{24}(15 + 0 - 3 - 6 + 18) = 1$

No. of $A_2 = \frac{1}{24}[(15\times1\times1) + 0 + (-1\times3\times1) + (-1\times-1\times6) + (3\times-1\times6)]$

$= \frac{1}{24}(15 - 3 + 6 - 18) = 0$

No. of $E = \frac{1}{24}[(15\times1\times2) + 0(-1\times3\times2) + 0 + 0)$

$= \frac{1}{24}(30 - 6) = 1$

No. of $T_1 = \frac{1}{24}[(15\times3\times1) + 0\ (-1\times-1\times3) + (-1\times6\times1) + (-1\times3\times6)]$

$= \frac{1}{24}(45 + 0 + 3 - 6 - 18)$

$= T_1$

No. of $T_2 = \frac{1}{24}[(15\times3\times1) + 0 + (-1\times-1\times3) + (-1\times-1\times6) + (3\times1\times6)]$

$= \frac{1}{24}(45 + 3 + 6 + 18) = 3$

So, $\Gamma = A_1 + A_2 + E + T_1 + 3T_2$

There are total molecular movement = 15

According to the character table of T_d

Total movement = $A_1 + E + T_1 + 3T_2$

Translation = T_2

Rotation = T_1

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vibrations = $A_1 + E + T_1 + 2T_2$

There are 9 modes of vibration is possible

The IR active vibrations are = $2T_2$

The Raman active vibrations are = $A_1 + E + 2T_2$

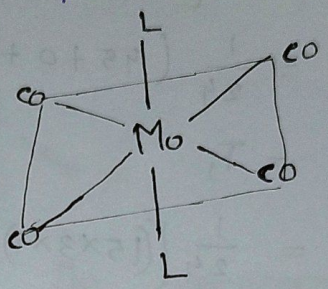
Therefore, there 6 vibration is IR active, and all 9 vibrations are Raman active.

Total band In IR → 2

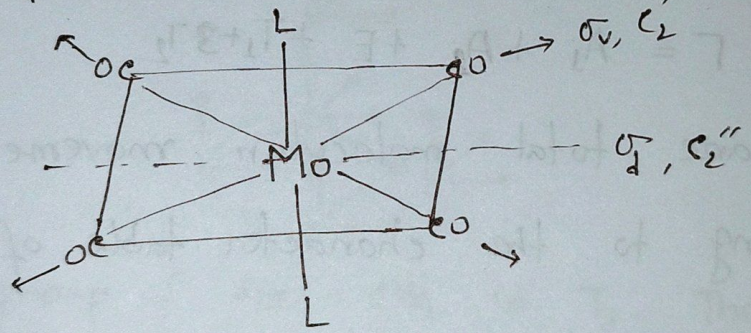
Total band In Raman → 4

⇒ Find the no. of terminal c-o stretching vibrations which are active in the infrared and Raman spectra of $Mo L_2(CO)_4$.

Structure,



Here, the total no. of c-o bond is 4. The basis of the representation is the set of the 4 arrows.



The four-arrow basis shown can be used to represent the carbonyl stretching vibrations. The point group of $Mo L_2(CO)_4$ is D_{4h}. The complete set of the character of the reducible representation is,

D_{4h}	E	$2C_4$	$C_2(=C_4^2)$	$2C_2'$	$2C_2''$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$
	4	0	0	2	0	0	0	4	2	0

According to the character table of C_{4h},

No. of $A_{1g} = \frac{1}{16}[(4\times1\times1) + 0 + 0 + (2\times2\times1) + 0 + 0 + 0 + (4\times1\times1) + (2\times2\times1) + 0]$

$= \frac{1}{16}(4 + 0 + 0 + 4 + 4 + 0 + 0 + 0 + 4 + 4 + 0)$

$= 1$

No. of $A_{2g} = 0$.

No. of $B_{1g} = 1$

No. of $B_{2g} = 0$

No. of $E_g = 0$

No. of $A_{1u} = 0$

No. of $A_{2g} = 0$

No. of $B_{1u} = 0$

No. of $B_{2u} = 0$

No. of $E_u = 1$

So, $\Gamma = A_{1g} + B_{1g} + E_u$

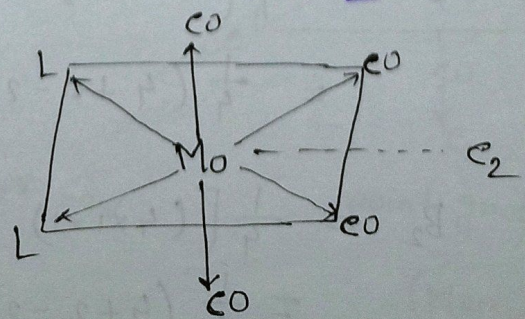
our basis only included stretching of the c-o bonds, we do not need to remove translations or rotation because we did not put them in when setting up the basis of the representation.

(1) The IR active stretching mode is E_u

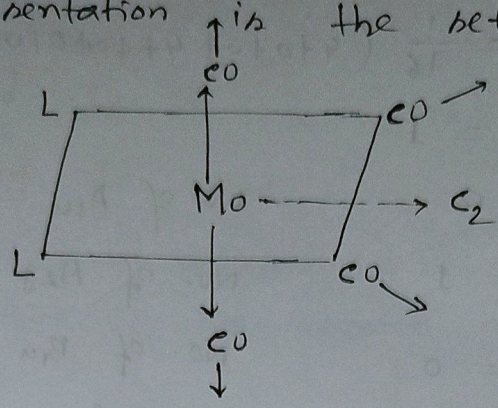
(2) The Raman active stretching mode is A_{1g}, B_{1g}

⇒ Find the no. of terminal c-o stretching vibrations which are active in the infrared and Raman spectra of the cis isomer of $(CO)_4 M L_2$

Structure:



Here, the total no. of CO bond is 4. The basis of the representation is the set of the 4 arrows.



The point group of the cis- $Mo L_2(CO)_4$ is C_{2v}.
The complete set of the character of the reducible representation is,

C_{2v}	E	C_2	σ	σ'
Γ	4	0	2	2

According to the character table of C_{2v},

No. of $A_1 = \frac{1}{4}[(4\times1\times1) + 0 + (2\times1\times1) + (2\times1\times1)]$

$= \frac{1}{4}(4+4) = 2$

No. of $A_2 = \frac{1}{4}[(4\times1\times1) + 0 + (2\times1\times-1) + (2\times1\times-1)$

$= \frac{1}{4}(4+0-2-2) = 0$

No. of $B_1 = \frac{1}{4}[(4\times1\times1) + 0 + (2\times1\times1) + (2\times1\times-1)]$

$= \frac{1}{4}(4+0-2+2) = 1$

No. of $B_2 = \frac{1}{4}[(4\times1\times1) + 0 + (2\times1\times-1) + (2\times1\times1)]$

$= \frac{1}{4}(4+2-2) = 1$

So, $\Gamma = 2A_1 + B_1 + B_2$

our basis is only included stretching of the c-o bonds, we do not need to remove translations or rotations because we did not put them in when setting up the basis of the representation.

(1) IR active stretching mode = $2A_1 + B_1 + B_2$
They belongs to x, y, z.

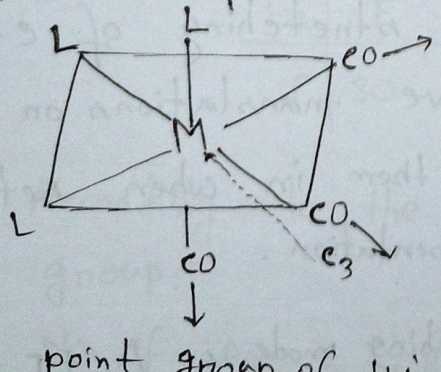
(2) Raman active stretching mode = $2A_1 + B_1 + B_2$
They belongs to x^2, y^2, z^2, xz, yz.

The all IR and Raman coincident to each other

→ 2020 (Fac-)

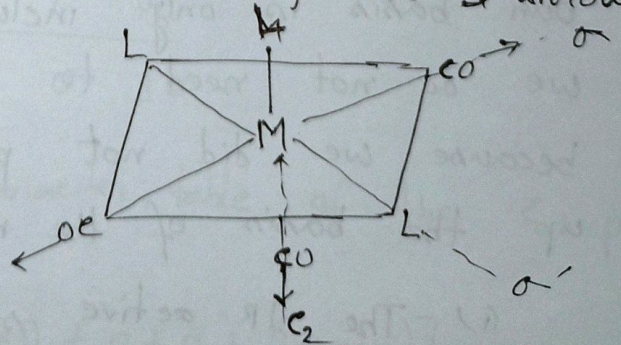
⇒ Find the no. of CO stretching vibrations which are active in the IR and Raman specta of a metal tri carbonyl of $ML_3(CO)_3$.

Here the total no. of CO bond is 3. The basis of the representation is the set of the 3 arrows.



point group of this C_{3V}

(Fac-isomer)



point gnou of this C_{2V}

(Mer-isomer)

The complete set of the chanacter of the reducible nepnesentation is,

C_{3v}	E	$2C_3$	$3\sigma_v$
Γ	3	0	1

According to the chanacter table of C_{3v}.

No. of $A_1 = \frac{1}{6}[(3\times1\times1)+0+(1\times3\times1)]$

$= \frac{1}{6}(3+3) = 1$

No. of $A_2 = \frac{1}{6}[(3\times1\times1)+0+(1\times3\times-1)]$

$= \frac{1}{6}(3-3) = 0$

No. of $E = \frac{1}{6}[(3\times2\times1)+0+0]$

$= \frac{1}{6}(6+0+0) = 1$

So, $\Gamma = A_1 + E$

C_{2v}	E	C_2	σ	σ'
	3	1	3	1

According to the chanacter table of C_{2v}.

No. of $A_1 = \frac{1}{4}(3+1+3+1) = 2$

No. of $A_2 = \frac{1}{4}(3+1-3-1) = 0$

No. of $B_1 = \frac{1}{4}(3-1+3-1) = 1$

No. of $B_2 = \frac{1}{4}(3-1-3+1) = 0$

So, $\Gamma = 2A_1 + B_1$

our basis is only included stnetching of c-o bonds we do not need to nemove tnanslations on notations because we did not put them in when setting up the basis of the nepnesentation.

(i) The IR active stnetching mode is $A_1 + E$

They belongs to the x, y, z axis.

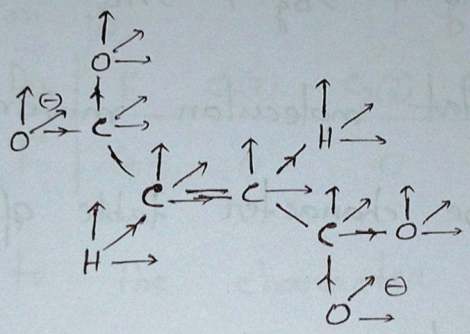
(ii) The Raman active stnetching mode is $= A_1 + E$

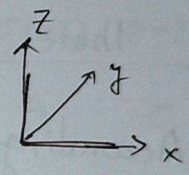
C_{3v}: 2 IR bands
2 Raman bans } coincident, $A_1 + E$

C_{2v}: 3 IR bands
2 Raman bands } coincident, $2A_1 + B_1$

Applications to molecular vibration test:

(1) Here N=10, The basis of the representation is the set of the $10 \times 3 = 30$ arrows.





The ionties in the xy plane. The C_2 axis is the z axis.

The point group of this ion is C_{2h}. The complete set of the character of the reducible representation is,

C_{2h}	E	C_2	i	σ_h
Γ	30	0	0	10

According to the character table of the C_{2h} point group:

No. of Ag $= \frac{1}{4}[(30 \times 1 \times 1) + 0 + 0 + (1 \times 1 \times 10)]$

$= \frac{1}{4}(30 + 10) = 10$

No. of $B_g = \frac{1}{4}[(30\times1\times1) + 0 + 0 + (10\times1\times-1)]$

$= \frac{1}{4}(30-10) = 5$

No. of $A_u = \frac{1}{4}[(30\times1\times1) + 0 + 0 + (10\times1\times-1)]$

$= \frac{1}{4}(30-10) = 5$

No. of $B_u = \frac{1}{4}[(30\times1\times1) + 0 + 0 + (10\times1\times1)]$

$= \frac{1}{4}(30+10) = 4$

So, $\Gamma = 10A_g + 5B_g + 5A_u + 10B_u$

There are total molecular movement is $= 30$

According to the character table of C_{2h},

Total movement $= 10A_g + 5B_g + 5A_u + 10B_u$

translation $= A_u + 2B_u$

Rotations $= A_g + 2B_g$

$\therefore$ vibrations $= 9A_g + 3B_g + 4A_u + 8B_u$

There are total 24 mode of vibration is possible.

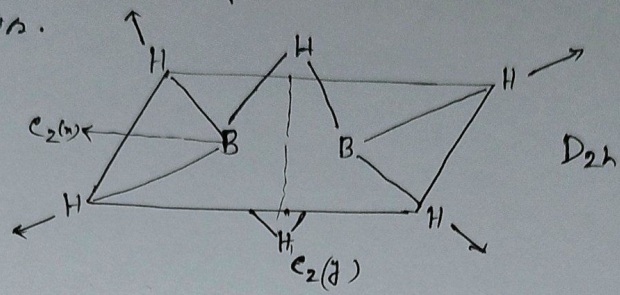
The IR active vibration are $= 4A_u + 8B_u$

They belongs to x, y, z axis.

The Raman active vibrations are $= 9A_g + 3B_g$

(2) See before.

(3) Here the total no. of B-H terminal bond is 4. The basis of the representation is the set of the 4 arrows.



The point group of this molecule is D_{2h}. The complete set of the character of the reducible representation is,

D_{2h}	E	$C_2(z)$	$C_2(y)$	$C_2(x)$	i	$\sigma(xy)$	σxz	σyz
Γ	4	0	0	0	0	4	0	0

According to the character table of D_{3h} point group.

No. of $Ag = \frac{1}{8}[(4\times1\times1) + 0+0+0+(4\times1\times1) + 0+0+0]$

$= 1$

No. of $B_1g = \frac{1}{8}[(4\times1\times1) + (4\times1\times1)] = 1$

$\Gamma = Ag + B_1g + B_{2u} + B_{3u}$.

our basis is only included stretching of B-H, we do not need to remove translation on rotation because we did not put them in when setting up the basis of the representation.

The IR active stretching mode is $B_{2u} + B_{3u}$

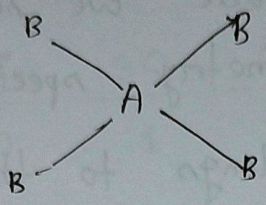
The Raman " " " " $A_{2g} + B_{1g}$

Chapter-6

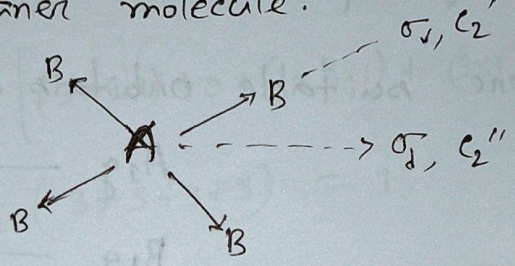
Applications to chemical bonding

(1) Find the hybrid orbitals of a central atom suitable for forming a set of square planer bonds. Use the D_{4h} character table.

Structure



Let us the 4 arrows representing the basis of the square planer molecule.



The point group of the square planer molecule is D_{4h}. The complete set of character of the reducible representation is,

D_{4h}	E	$2C_4$	$C_2(=C_4^2)$	$2C_2'$	$2C_2''$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$
Γ	4	0	0	2	0	0	0	4	2	0

According to the character table of D_{4h} point group.

No. of $A_{1g} = \frac{1}{16}[(4\times1\times1) + 0 + 0 + (2\times2\times1) + 0 + 0 + 0 + (4\times1\times1) + (2\times2\times1) + 0]$

$= \frac{1}{16}(4+4+4+4) = 1$

No. of $A_{2g} = 0$, No. of $B_{1g} = 1$, No. of $B_{2g} = 0$

No. of $E_g = 0$, No. of $A_{1u} = 0$, No. of $A_{2u} = 0$

No. of $B_{1u} = 0$, No. of $B_{2u} = 0$, No. of $E_u = 1$

So we can see that this reduces to

$= A_{1g} + B_{1g} + E_u$

From the character table we see that s or dz^2 belongs to the symmetry species A_{1g}.

dx^2-y^2 orbital belongs to the B_{1g} symmetry species and P_x and P_y orbitals belongs to the E_u symmetry species.

Therefore suitable orbitals are:

$A_{1g} \longrightarrow$ s or dz^2

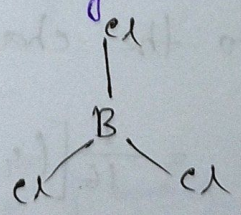
$B_{1g} \longrightarrow dx^2-y^2$

$E_u \longrightarrow P_x$ and P_y

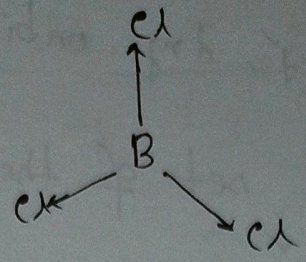
Hence a set of hybrid orbitals are dsp^2.

(2) Find the hybrid orbitals of a central atoms suitable for forming a set of BCl_3 bonds.

Structure:



Let us the 3 arrows representing the basis of the BCl_3 molecule.



The point group of the BCl_3 molecule is D_{3h}. The complete set of character of the reducible representation is,

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ	3	0	1	3	0	1

According to the character table of D_{3h} point group:

No. of $A_1' = \frac{1}{12}\left[(3\times1\times1) + 0 + (1\times3\times1) + (3\times1\times1) + 0 + (1\times3\times1)\right]$

$= \frac{1}{12}(3+3+3+3) = 1$

No. of $A_2' = 0$, No. of $A_2'' = 0$

No. of $E' = 1$ No. of $E'' = 0$

No. of $A_1'' = 0$

so we get, $\Gamma = A_1' + E'$

From the character table we see that, s or d_{z^2} belongs to the symmetry species A_1'.

(p_x, p_y) or $(d_{x^2-y^2}, d_{xy})$ belongs to the symmetry species E'.

In B, there has no energetically d-orbitals. So

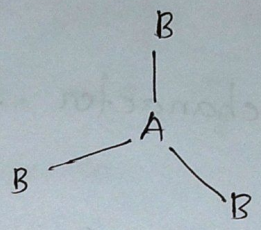
we can neglet $d_{x^2-y^2}$, d_{xy} onbitals for hybridiz

So, the most likely set of the hybrids onbitals ~~onbitals~~ are sp

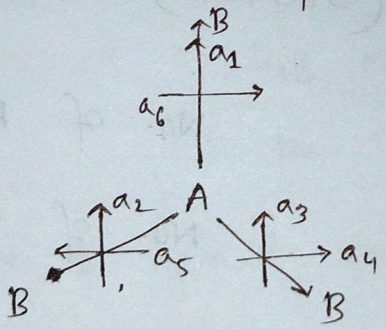
This is true for all trigonal set of bonds.

(3) Find the onbitals suitable for AB_3 (BCl_3) to form a π-bonding.

Structure:



Let us the six-arrows nepresentating the six-p onbitals as a basis of the AB_3 molecules.



There are in two sets a_1, a_2, a_3 - the out of the plane and a_4, a_5, a_6 - the in plane set.

The point group of the AB_3 molecule is D_{3h}. The complete set of chanacten of the neducible nepnesentation is,

For the out of plane set,

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ_1	3	0	-1	-3	0	1

For the in a plane, set,

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ_2	3	0	-1	3	0	-1

According to the chanacter table of D_{3h} point group.

For in a plane

No. of A_1' $= \frac{1}{12}\left[(3\times1\times1) + 0 + (1\times-3\times1) + (3\times1\times1) + (1\times3\times-1)\right]$

$= \frac{0}{12} = 0$

No. of A_2' $= \frac{1}{12}\left[(3\times1\times1) + 0 + (-1\times3\times-1) + (3\times1\times1) + 0 + (-1\times3\times-1)\right]$

$= \frac{1}{12}(3+3+3+3) = 1$

No. of E' $= 1$

No. of A_1'' $= 0$

No. of A_2'' $= 0$

No. of E'' $= 0$

So we get, Γ_2 (in a plane) $= A_2' + E'$

For out of plane

No. of A_1' $= 0$

No. of A_2' $= 0$

No. of E' $= 0$

No. of A_1'' $= 0$

No. of A_2'' $= 1$

No. of E'' $= 1$

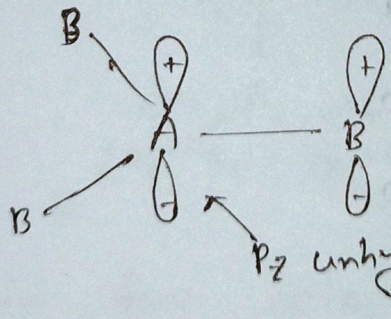
So, we get, Γ(out of plane) = $A_2'' + E''$

From the chanacter table we see that,

out of plane: P_z, (d_{xz}, d_{yz}) onbitals ane belongs to the A_2'' and E_2'' nepnesentation nespectively.

In a plane: (P_x, P_y) on ($d_{x^2-y^2}$, d_{xy}) onbitals ane belongs to the E'' nepnesentation.

Since, A (Boron) has no enegetically available d onbital. So we can neglet d_{xz}, $d_{x^2-y^2}$, d_{xy}, d_{yz} for the π-bonding. Now since P_x and P_y onbitals ane already used for hybnidization, only P_z onbital is available for π-bonding.

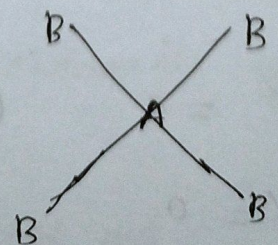


P_z unhybnidized onbitals.

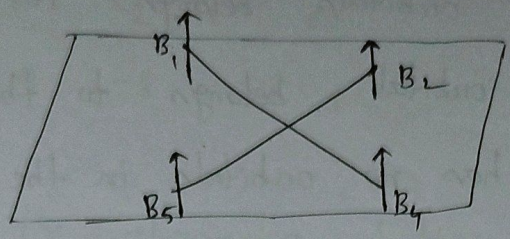
(4) Find the onbitals suitable for the out of plane π-bonding in a square plane molecule.

→ hybnidization → dsp^2 ✓

Structune:



Let us the 4-arrows nepresenting the 4-onbitals as a basis of the AB_4 molecule (out of plane)



arrows are out of plane

The point gnoup of the squane planen molecule is D_{4h}. The complete set of the chanacten of the neducible nepnesentation is,

D_{4h}	E	$2C_4$	$C_2 (= C_4^2)$	$2C_2'$	$2C_2''$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$
Γ	4	0	0	-2	0	0	0	-4	2	0

According to the chanacten table of the D_{4h} point gnoup.

No. of $A_{1g} = \frac{1}{16}\left[(4\times1\times1) + 0 + 0 + (-2\times2\times2) + 0 + 0 + 0 + (-4\times1\times1) + (2\times2\times1)\right]$

$= \frac{1}{16}(4 - 4 + 4 - 4) = 1$

No. of $A_{2g} = 0$ No. of $B_{1u} = 0$

$B_{1g} = 0$, $B_{2g} = 0$.

No. of $Eg = 1$ No. of $B_{2u} = 1$

No. of $A_{1u} = 0$ No. of $Eu = 0$

No. of $A_{2u} = 1$

The inneducible nepnesentation of out of plane,

So, $\Gamma = Eg + A_{2u} + B_{2u}$

From the character table we see that,

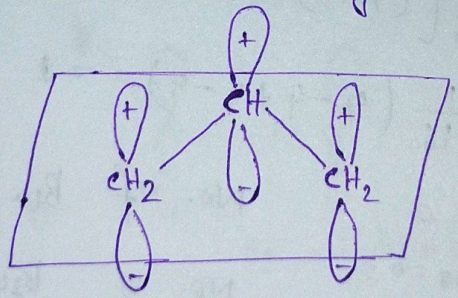
d_{xy}, d_{yz} orbitals belongs to the E_g representation.

P_z orbital belongs to the A_{2u} representation.

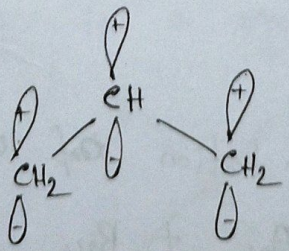
There has no orbitals in the B_{2u} representation.

Therefore, d_{xy}, d_{yz} and P_z orbitals are suitable for the out of plane π-bonding in a square planer molecule.

⇒ How many symmetries of the LCAO π-molecular orbitals of the open chain C_3 system. Use the C_{2v} character table. How many different energy levels will there be in the symmetry system?



Let us 3-p orbitals representing the basis of this molecules.



The point group of these molecule is C_{2v}. The

complete set of chanacter of the reducible nepre-sentation is,

C_{2v}	E	C_2	σ	σ'
Γ	3	-1	-3	1

Aecording to the chanacter table of C_{2v} point group.

No. of $A_1 = \frac{1}{4}[(3\times1\times1) + (-1\times1\times1) + (-3\times1\times1) + (1\times1\times1)]$

$= \frac{1}{4}(3-1-3+1) = 0$

No. of $A_2 = \frac{1}{4}[(3\times1\times1) + (-1\times1\times1)\ (-3\times1\times-1) + (-1\times1\times1)]$

$= \frac{1}{4}(3-1+3-1) = 1$

No. of $B_1 = 0$.

No. of $B_2 = \frac{1}{4}[(3\times1\times1) + (-1\times-1\times1)\ (3\times1\times-1) + (1\times1\times1)]$

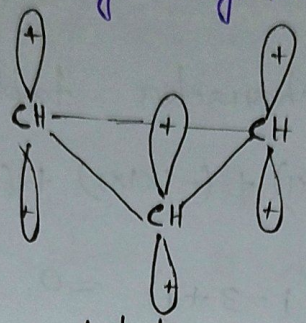
$= \frac{1}{4}(3+1+3+1) = 2$

So we can nee that this is neduced to,

$$\Gamma = A_2 + 2B_2$$

From the inineducible nepnesentation we nee that, there han 3 -onbitals, All of different energy.

⇒ How many symmetries of the LCAO π-molecular orbitals of this system. Use the D_{3h} character table. How many different energy levels will there be in the symmetry system?



cyclopropenyl radical.

Let us three 3p-orbitals representing the basis of the cyclopropenyl radical,

Figure

The point group of these molecule is C_{3h}. The complete set of the character of the reducible representation is,

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
	3	0	-1	-3	0	1

According to the character table of D_{3h} point group

No. of $A_1' = \frac{1}{12}[(3\times1\times1) + 0 + (3\times-1\times1) + (-3\times1\times1) + 0 + (3\times1\times1)]$

$= \frac{1}{12}[3+3-3-3) = 0.$

No. of $A_2' = \frac{1}{12}[(3\times1\times1) + 0 + (3\times-1\times-1) + (-3\times1\times1) + 0 + (3\times-1\times1)]$

$= \frac{1}{12}[3+0+3-3-3] = 0.$

No. of $E = 0$

No. of $A_1'' = \frac{1}{12}[(3\times1\times1) + 0 + (3\times-1\times1) + (-3\times1\times-1) + 0 + (3\times1\times-1)]$

$= \frac{1}{12}(3-3+3-3) = 0.$

No. of $A_2'' = \frac{1}{12}[(3\times1\times1) + 0 (3\times-1\times-1) + (-3\times1\times-1) + 0 + (3\times1\times1)]$

$= \frac{1}{12}(3+3+3+3) = 1$

No. of $E'' = \frac{1}{12}[(3\times2\times1) + 0 + 0 + (-3\times1\times-2) + 0 + 0]$

$= \frac{1}{12}(6+0+0+6) = 1$

So, we can see that this is reduced to,

$$\Gamma = A_2'' + E''$$

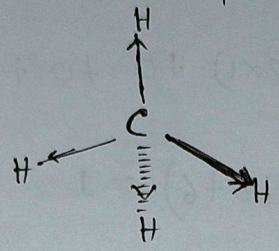
From this representation, we see that there has 3-orbitals. The molecular orbitals consists of a doubly degenerate pair (E'') and one singly degenerate orbital (A''). This result tell us nothing about the energy difference between the A_2'' and E'' orbitals.

2020

Set up the molecular orbital correlation diagram for the CH_4 molecule, considering the 2s and 2p orbitals of carbon and the 1s orbital of each hydrogen atom.

The structure of the CH_4 molecule is, tetrahedral,

Structure:



CH_4 molecules belongs to the T_d point group.

From the point group we see that,

1s and 2s orbitals belongs to A_1 representation.

$2P_x$, $2P_y$ and P_z " " " T_2 "

By taking the four (4) C-H bond as the basis, we can calculate the group orbitals for the four (4) H atoms, The complete set of character of the reducible representation is,

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$
Γ	4	1	0	0	2

According to the character table of T_d point group,

No. of $A_1 = \frac{1}{24}\left[(4\times1\times1) + (1\times8\times1) + 0 + 0 + (2\times6\times1)\right]$

$= \frac{1}{24}(4+8+12) = 1$

No. of $A_2 = \frac{1}{24}[(4\times1\times1) + (1\times1\times8) + 0 + 0 + (2\times-1\times8)] = 0$

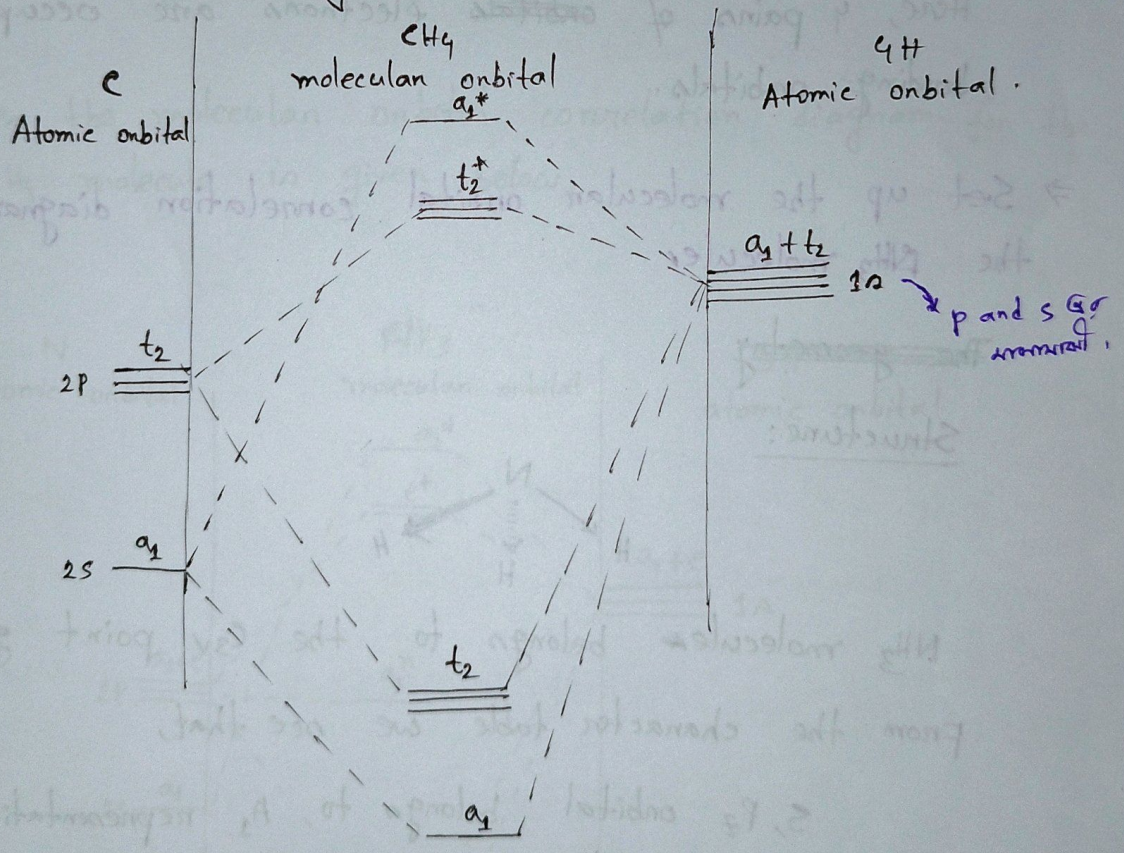
No. of $E = 0$

No. of $T_1 = 0$

No. of $T_2 = 1$

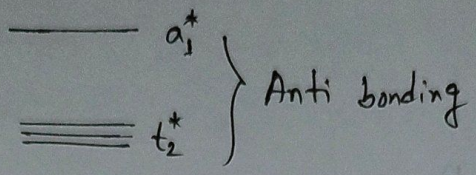
So we get, $\Gamma = A_1 + T_2$

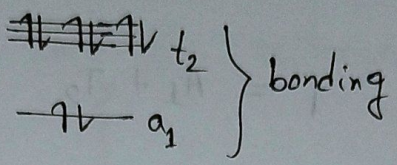
So, the molecular orbital correlation diagram for the CH_4 molecule is given below.



In CH_4, carbon has 4 valance electrons, in its valance shell. and 4-hydrogen atoms have 4 valance electrons. So, putting these $(4+4) = 8$ electrons in

the molecular orbitals as follows.





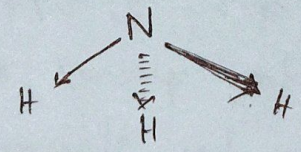
Bond order $= \frac{8-0}{2} = 4$

Here, 4 pairs of ~~orbitals~~ electrons are occupy bonding orbitals.

⇒ Set up the molecular orbital correlation diagram for the NH_3 molecule.

~~The geometry~~

Structure:



NH_3 molecule belongs to the C_{3v} point group. From the character table we see that,

s, p_z orbital belongs to A_1 representation.

p_x, p_y orbitals belongs to E representation.

By taking the ~~four~~ three(3) N-H bond as the basis, we can calculate the group orbitals for the three (3) H atoms, The complete set of character of the reducible representation

in,

C_{3v}	E	$2C_3$	$3\sigma_v$
Γ	3	0	1

According to the chanacter table of C_{3v} point ~~table~~ group,

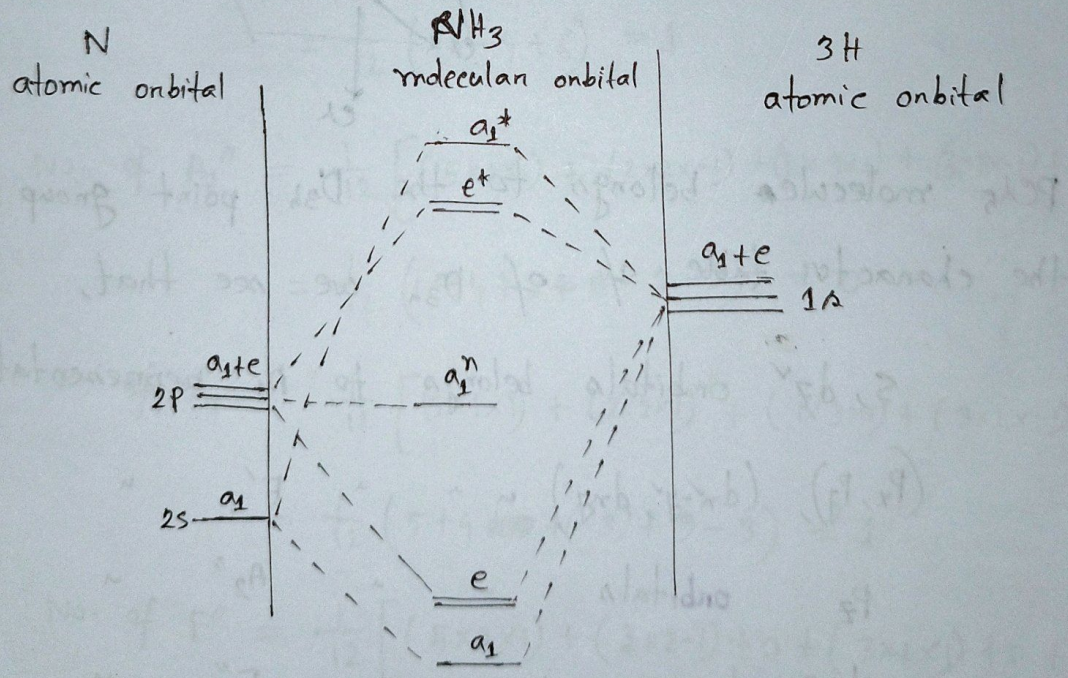
No. of $A_1 = \frac{1}{6}[(3\times1\times1) + 0 + (3\times1\times1)] = 1$

No. of $A_2 = \frac{1}{6}[(3\times1\times1) + 0 + (1\times(-1)\times3) = 0$

No. of $E = \frac{1}{6}[(3\times2\times1) + 0 + 0] = 1$

So, we get $\Gamma = A_1 + E$

So, the molecular onbital correlation diagnam for the CH_4 molecule is given below,



In NH_3, Nitrogen has 5 valance electrons, and 3-H atoms have 3-valance electrons. So putting these $(5+3) = 8$

electrons in the molecular orbitals as follows,

a_1^* } antibonding
e^* }

Bond order =

$\frac{1}{2}$ (No. of b.e - No. of A.b.e)

$= \frac{1}{2}(6-0)$

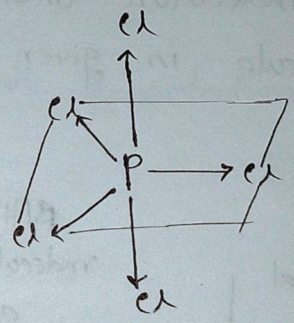
$= 3$

a_1^n } non-bonding

e } bonding
a_1 }

⇒ Set up the molecular orbital correlation diagram for the PCl_5 molecules.

Structure:



PCl_5 molecule belongs to the D_{3h} point group. From the character table of of D_{3h} we see that,

s, d_{z^2} orbitals belongs to A_1 representation.

(p_x, p_y), $(d_{x^2-y^2}, d_{xy})$ " " " E' "

p_z orbitals " " A_2'' "

d_{xz}, d_{yz} " " " E'' "

By taking the Five (5) P-Cl bond as the basis we can calculate the group orbitals for the five(5)

Cl atoms, The complete set of the chanacter of the reducible nepnesentation is,

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ	5	2	1	3	0	3

Aceonding to the chanacter table of D_{3h} point gnoup.

No. of $A_1' = \frac{1}{12}\left[(5\times1\times1) + (2\times2\times1) + (3\times1\times1) + (3\times1\times1) + 0 + (3\times3\times1)\right]$

$= \frac{1}{12}(5+4+3+3+0+9) = 2$

No. of $A_2' = \frac{1}{12}\left[(5\times1\times1) + (2\times2\times2) + (3\times1\times-1) + (3\times1\times1) + 0 + (3\times3\times-1)\right]$

$= \frac{1}{12}(5+4+3-3-9) = 0.$

No. of $E' = \frac{1}{12}\left[(5\times2\times1) + (2\times2\times-1) + 0 + (3\times1\times2) + 0 + 0\right]$

$= \frac{1}{12}(10-4+6) = 1$

No. of $A_1'' = \frac{1}{12}\left[(5\times1\times1) + (2\times2\times-1) + (1\times3\times1) + (3\times1\times-1) + 0 + (3\times3\times1)\right]$

$= \frac{1}{12}(5+4+3+0-3-9) = 0$

No. of $A_2'' = \frac{1}{12}\left[(5\times1\times1) + (2\times2\times1) + (1\times3\times1) + (3\times1\times-1) + 0 + (3\times3\times1)\right]$

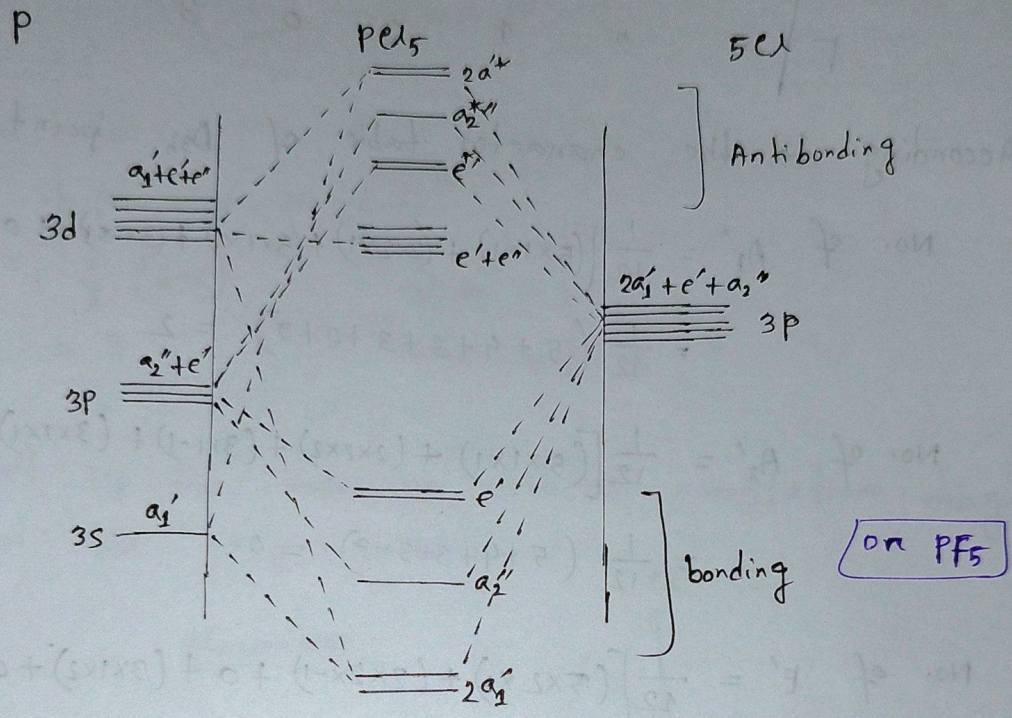
$= \frac{1}{12}(5+4+(-3)+9-3) = 1$

No. of $E'' = \frac{1}{12}\left[(5\times2\times1) + (3\times2-1) + 0 + (3\times2\times1) + 0 + 0\right]$

$= \frac{1}{12}(10-4-6) = 0.$

So, we get, $\Gamma = 2A_1' + E' + A_2''$

So, the molecular orbital correlation diagram for the ~~PCl5~~ molecule is given below.



~~In PCl5, P has 5 valance electrons and Cl/F has~~

In PCl_5 there has total ten electrons in the B.M.O orbita, there has no electrons in the antibonding orbitals.

$2a_1'^*$
$a_2''^*$
e'^*
} Antibonding

$e'+e''$ } non bonding

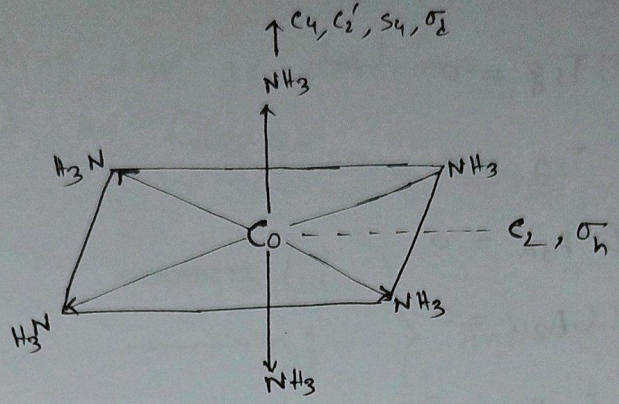
Bond order $= \frac{10-0}{2}$

$= 5$

~~↑↓~~ e'
~~↑↓~~ a_2''
~~↑↓~~ $2a_1'$
} bonding

⇒ Set up the molecular orbitals correlation diagram for the $[Co(NH_3)_6]^{3+}$ complex.

Structure:



$[Co(NH_3)_6]^{3+}$ complex is belongs to the O_h point group.

From the point table O_h, we see that,

$3d_{x^2-y^2}$ and $3d_{z^2}$ belongs to E_g representation.

$3d_{xy}, 3d_{yz}, 3d_{zx}$ " " T_{2g} "

4S orbitals " " A_{1g} "

4p " " T_{1u} "

By taking the Six(6) $Co-NH_3$ bond as the basis we can calculate the group orbitals for the Six(6) NH_3.

The complete set of the character of the reducible representation is,

O_h	E	$8C_3$	$6C_2$	$6C_4$	$3C_2' = (C_4^2)$	i	$6S_4$	$8S_6$	$3\sigma_h$	$6\sigma_d$
	6	0	0	2	2	0	0	0	4	2

According to the character table of O_h point group.

No. of $A_{1g} = \frac{1}{48}[(6\times1\times1) + 0 + 0 + (6\times2\times1) + (3\times2\times1) + 0 + 0 + 0 + (4\times3\times1) + (2\times6\times1)$

$= 1$

No. of $A_{2g} = 0$

No. of $E_g = 1$

No. of $T_{1g} = 0$

No. of $T_{2g} = 0$

No. of $A_{1u} = 0$

No. of $A_{2u} = 0$

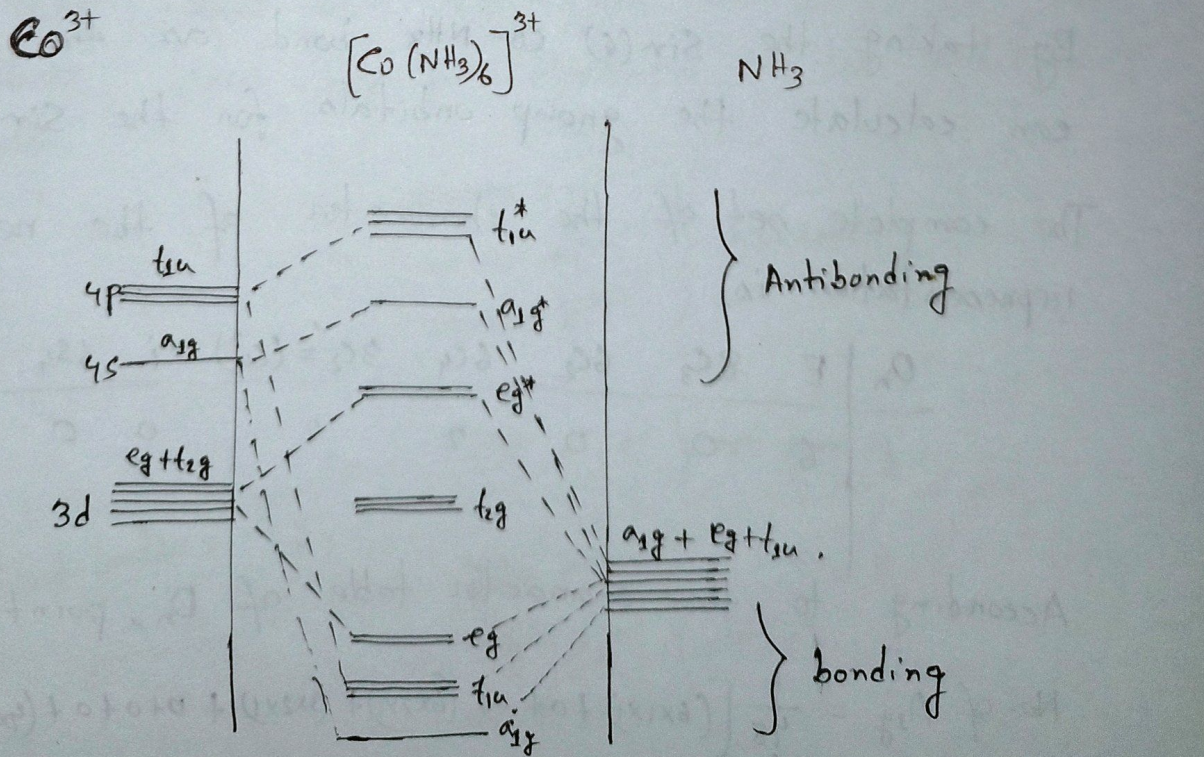
No. of $E_u = 0$

No. of $T_{1u} = 1$

No. of $T_{2u} = 0$

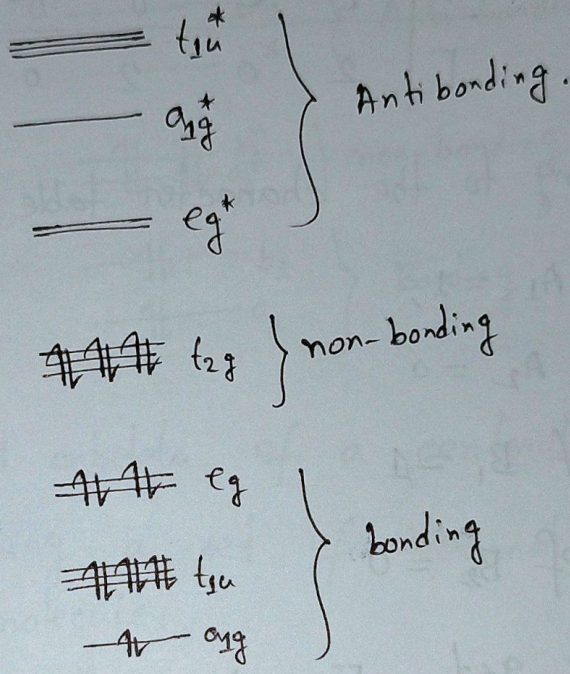
So we get, $\Gamma = A_{1g} + E_g + T_{1u}$

So the molecular orbital correlation diagram for the $[Co(NH_3)_6]^{3+}$ complex is given below,



The complex $Co(NH_3)_6]^{3+}$ has eighteen electrons in the onbitals under consideration.

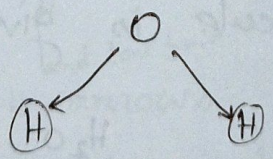
So putting the 18 electrons in the molecular onbitals as follows.



$$B.O = \frac{12-0}{2} = 6$$

⇒ Set the comelation diagnam for H_2O molecule.

Structure:



H_2O molecule belongs to the C_{2v} point group.

From the C_{2v} point group we see that,

s and P_z onbitals belongs to A_1 nepnesenta-tion.

P_x " " " B_1 "

P_y " " " B_2 "

By taking the two (2) O–H bond as the basis we can calculate the group orbitals for the two (2) H-atoms, the complete set of the character of the reducible representation is,

C_{2v}	E	C_2	σ	σ'
Γ	2	0	2	0

According to the character table of C_{2v} point group.

No. of $A_1 = 1$,

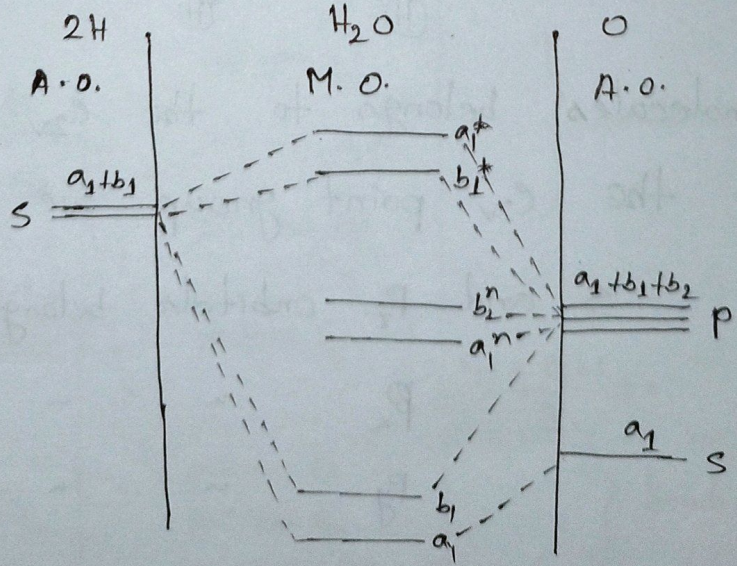
No. of $A_2 = 0$

No. of $B_1 = 1$

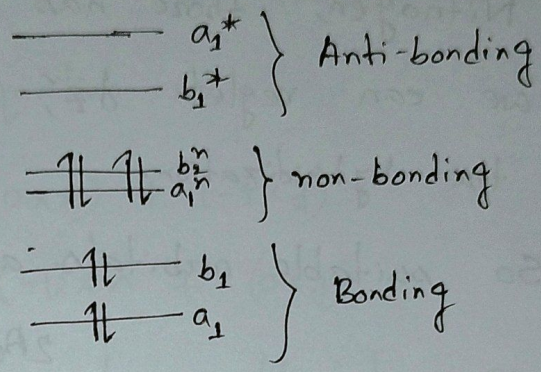
No. of $B_2 = 0$

So we get, $\Gamma = A_1 + B_1$

So the molecular orbital correlation diagram for the H_2O molecule is given below.



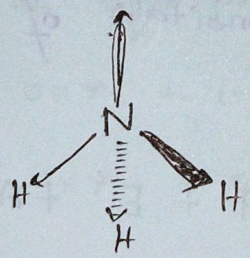
In H_2O, oxygen has 6 valance electrons, in its valance shell, and 2-H atoms have 2-valance electrons. So putting these $(6+2) = 8$ electrons in the molecular orbitals as follows.



Bond order $= \frac{4-0}{2}$
$= 2$

⇒ Find the hybrid orbitals of a central atom suitable for forming a set of trigonal pyramidal bond. For NH_3 molecule:

Structure:



and lone pair

Let us the 3 N-H narrows representing the basis of the trigonal pyramidal molecules (NH_3)

The point group of the NH_3 molecule is C_{3V}. The complete set of character of the reducible representation is,

C_{3v}	E	$2C_3$	$3\sigma_v$
Γ	3	0	1

This is reduced to, $\Gamma = 2A_1 + E$

From the chanacter table we see that,

$S, 2P_z, d_{z^2}$ onbitals belongs to the A_1 nepnesentation.

(P_x, P_y), $(d_{x^2-y^2}, d_{yz})$ " " " " E "
d_{xy}, d_{zx},

In Nitnogen, there has no enengetically d-onbitals So we can neglet d_{z^2}, $d_{x^2-y^2}$, d_{xy}, d_{yz}, d_{zx} onbitals fnom the hybnidization.

So suitable onbitals ane,

$2A_1$ — S, P_z

E — P_x, P_y

Hence a set of sp^3 hybnid onbitals.

⇒ Find the hybnid onbitals of a centenal atom in PCl_5 molecule (D_{3h})

$$\Gamma = 2A' + E' + A_2''$$

So suitable onbitals ane,

$2A' \longrightarrow S, d_{z^2}$

$E' \longrightarrow P_x, P_y$

$A_2'' \longrightarrow P_z$

Hence a set of sp^3d hybnid onbitals.

Final: 2020

3(c) Setup the matrixes which will perform the following transformation:

(A) $\begin{pmatrix} x \\ y \end{pmatrix}$ to $\begin{pmatrix} -y \\ -x \end{pmatrix}$

The full set of equation is,

$$x = (0 \times x) + (-1 \times y)$$

$$y = (-1) \times x + (0 \times y)$$

The matrices can be written down as $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

(b) $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ to $\begin{pmatrix} -y \\ x \\ -z \end{pmatrix}$

The full set of equation is,

$$x = 0 \times x + (-1) \times y + 0 \times z$$

$$y = 1 \times x + 0 \times y + 0 \times z$$

$$z = 0 \times x + 0 \times y + (-1) \times z)$$

The matrices can be written down by the inspection as, $\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

correlation diagram of BCl_3:

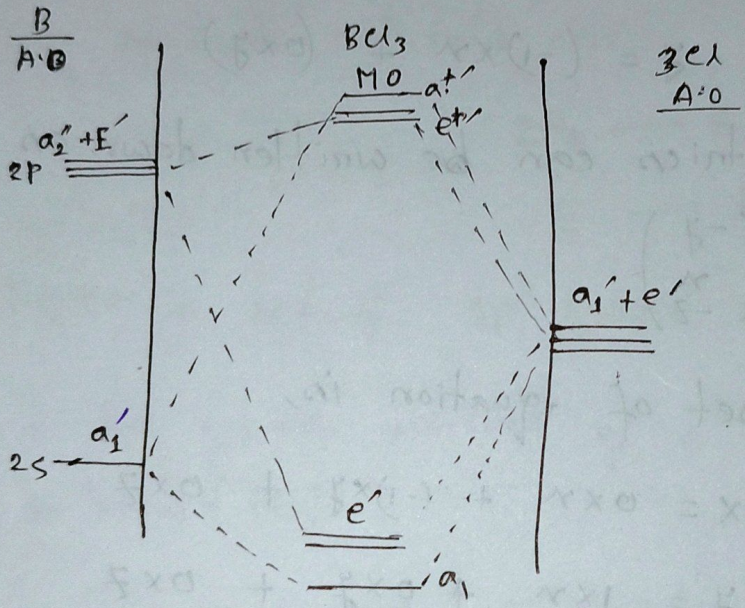
D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ	3	0	1	3	0	1

$$\Gamma = A_1' + E'$$

$2S \longrightarrow A_1'$

$2P_x, 2P_y \longrightarrow E'$

$2P_z \longrightarrow A_2''$



$B.O = \frac{6-0}{2} = 3.$

a_1' } Antibonding
e' }

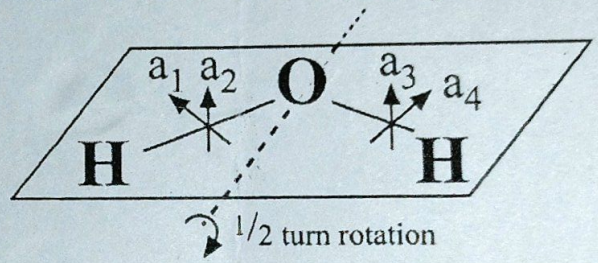
bond order $= \frac{6-0}{2}$

$= 3$

e' } bonding
a_1 }

Department of Chemistry, Jahangirnagar University
Course Title: Molecular Symmetry and Group Theory
Course no.: Chem. 324; Marks: 10, Time: 45 min.

1. Generate a matrix representation of the two fold rotation for H_2O molecule, using the four arrows shown as the basis (N. B. a_2 and a_3 are perpendicular to the plane, a_1 and a_2 are in it). Use the same four arrows as the basis to generate a matrix representation of the operation of reflection in the plane of the molecule. Also find the characters of these matrix representations. 3



2. Set up the molecular orbital correlation diagram for the CH_4 molecule, considering the 2*s* and 2*p* orbitals of carbon and the 1*s* orbital of each hydrogen atom. Also find the hybrid orbitals of carbon atom suitable for forming the four C–H bonds in CH_4 molecule (the Character Table of T_d point group is given below for your convenience). 7

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$		
A_1	1	1	1	1	1		$x^2 + y^2 + z^2$
A_2	1	1	1	-1	-1		
E	2	-1	2	0	0		$(2z^2 - x^2 - y^2, x^2 - y^2)$
T_1	3	0	-1	1	-1	(R_x, R_y, R_z)	
T_2	3	0	-1	-1	1	(x, y, z)	

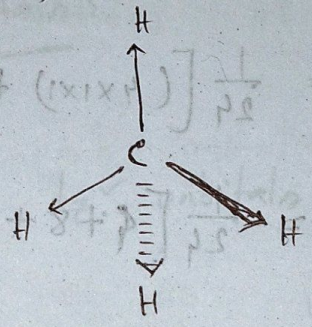
9.5/10

Department of Chemistry, JU
Name: Nazmul Hasan
Class Roll: 196 Session: 18-19
Course: 324 Date: 8-12-21

Ans to the Q. No. 2

Structure:

The structure of the methane (CH_4) is tetrahedral,



Let us, take the four (4) C–H bond as a basis of the CH_4 molecule.

CH_4 molecules is belongs to the T_d point group. The complete set of character of the reducible representation of CH_4 molecule is:

p.T.o.

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$
Γ	4	1	0	0	2

From the character table of the T_d point group,

No. of $A_1 = \frac{1}{24}\left[(4\times1\times1) + (1\times8\times1) + 0 + 0 + (2\times6\times1)\right]$

$= \frac{1}{24}(4+8+12) = 1$

No. of $A_2 = \frac{1}{24}\left[(4\times1\times1) + (1\times8\times1) + 0 + 0 + (2\times6\times-1)\right]$

$= \frac{1}{24}\left[4+8+0+0-12)\right)$

$= 0$

No. of $E = \frac{1}{24}\left[(4\times2\times1) + (1\times8\times-1) + 0 + 0 + (6\times2\times0)\right]$

$= \frac{1}{24}(8-8) = 0$

No. of $T_1 = \frac{1}{24}\left[(4\times3\times1) + 0 + 0 + 0 + (6\times2\times-1)\right]$

$= \frac{1}{24}(12+0+0+0+0-12)$

$= 0$.

No. of $T_2 = \frac{1}{24}[(3\times4\times1) + 0 + 0 + 0 + (2\times6\times1)]$

$= \frac{1}{24}(12 + 0 + 0 + 0 + 12)$

$= 1$

So the irreducible representation is,

$\Gamma = A_1 + T_2$

For hybrid orbitals:

S or d_{z^2} orbitals belongs to the A_1 representation,

P_x, P_y, P_z orbitals belongs to the T_2 representation.

Since there has no energetically favourable d-orbital in the carbon atom, so we can neglect the d_{z^2} from the hybridization.

So, $A_1 \longrightarrow S$

$T_2 \longrightarrow P_x, P_y, P_z$.

So the hybridization is, sp^3.

For correlation diagram:

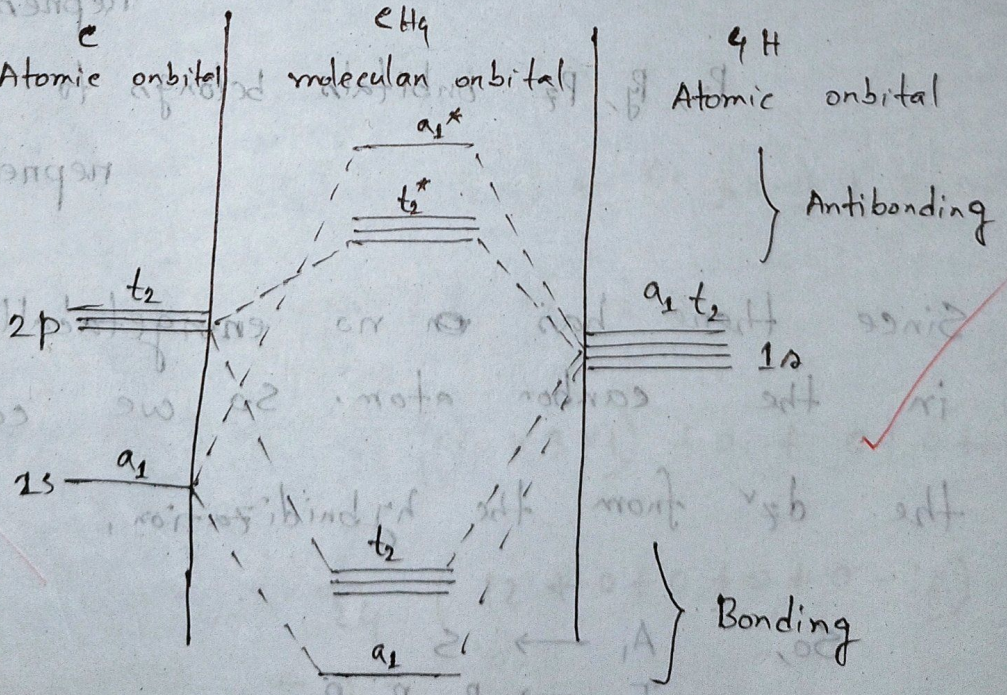
we have,

$$\Gamma = A_1 + T_2$$

S on d_{z^2} orbitals belongs to A_1 representation,

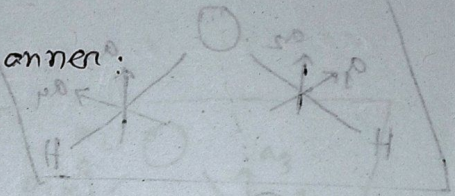
P_x, P_y, P_z orbitals belongs to T_2 representation;

So the correlation diagram of CH_4 molecule is:



here, the total no. of electrons is eight. four (4) electrons from the outermost onbitals of canbon atom and the foun electrons from the 4H atoms.

Therefone the electrons ame distributed by following manner:



a_1^*

t_2^* } antibonding

t_2

a_1 } Bonding.

Bond onden = $\frac{1}{2}$ (No. of electrons in bonding - No. of electrons in antibonding)

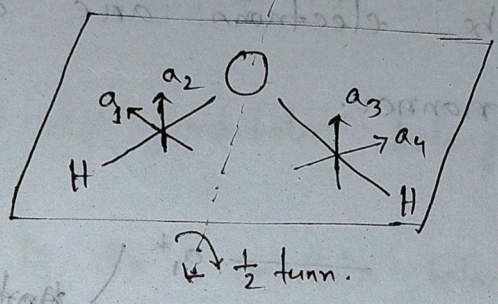
$= \frac{1}{2}(8-0)$

$= 4$

Bond onden is foun (4)

Ans to the Q.No.-1

For the two fold rotation of H_2O molecules,



a_1, a_3 are perpendicular

a_1, a_4 are plane.

$\frac{1}{2}$ turn.

By the $\frac{1}{2}$ turn rotation, a_1, a_2, a_3, a_4 becomes,

$$\begin{cases} a_1 = a_4 \\ a_2 = -a_3 \\ a_3 = -a_2 \\ a_4 = a_1 \end{cases}$$

The matrix representation of these,

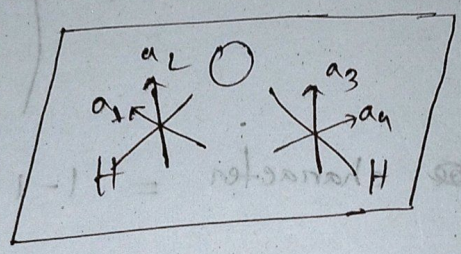
$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \begin{pmatrix} a_4 \\ -a_3 \\ -a_2 \\ a_1 \end{pmatrix}$$

The matrix is,

$$M = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

character = 0.

For in a plane:



a_1, a_4 in a plane

a_2, a_3 out of plane.

By reflection in the plane:

$$a_1 = a_1$$

$$a_2 = -a_2$$

$$a_3 = -a_3$$

$$a_4 = a_4$$

P.T.O.

The matrix representations of these,

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \begin{pmatrix} a \\ -a_2 \\ -a_3 \\ a_4 \end{pmatrix}$$

The matrix is, $M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

The character $= 1 - 1 - 1 + 1 = 0$.

The character of the matrix;

when C_2 operation is done $= 0$

when plane of reflection $= 0$.

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Department of Chemistry, JU

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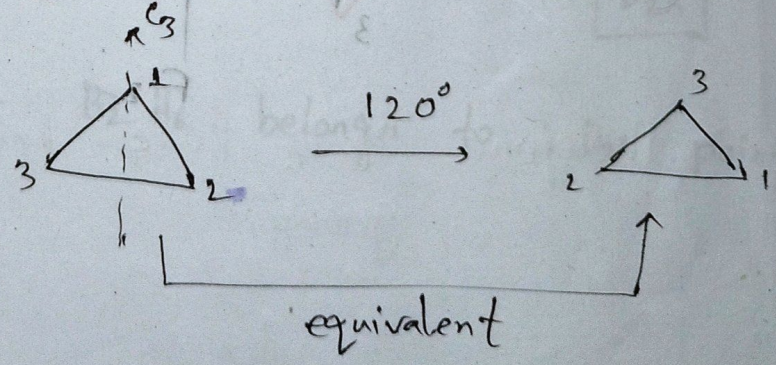
Class Roll: 196 Session: ~~9-11-22~~

Course: 324 Date: ~~9-11~~-22

Ans to the Q.No. 1

Symmetry operation: A symmetry operation is a movement of a body such that after the movement has been carried out every point of the body is coincident with an equivalent point of the body in its original orientation.

Ex:



(ii) Center of symmetry:

A molecule possess a center of symmetry if every atom when moved in straight line through the center and equal distance of the other side of the center, encounter a similar atom.

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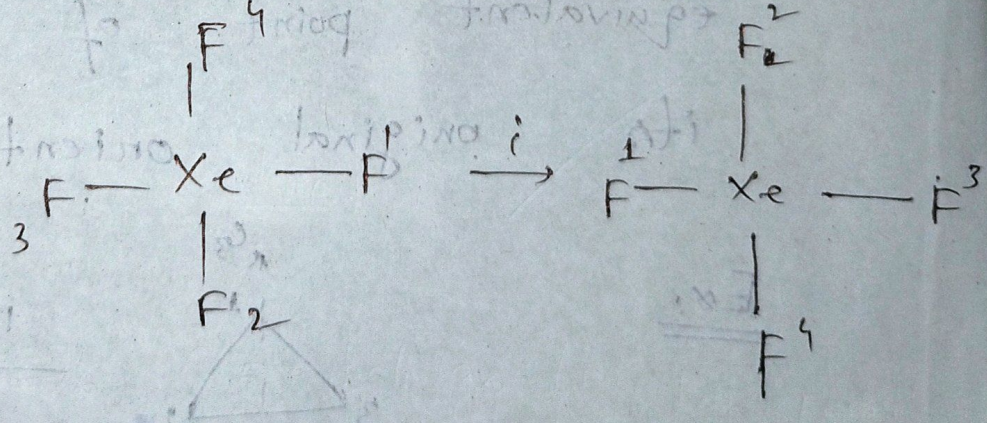
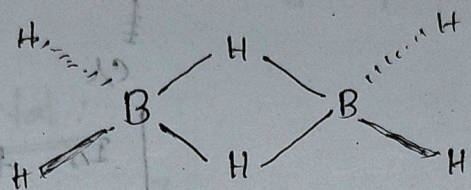
Ex:

XeF_4



Ans to the Q.N. (b)

Determination of the point group of B_2H_6



Is this molecule belongs to special point group?

No

C_n?

yes

$n = 2$

$C_2 \perp nC_2$

yes

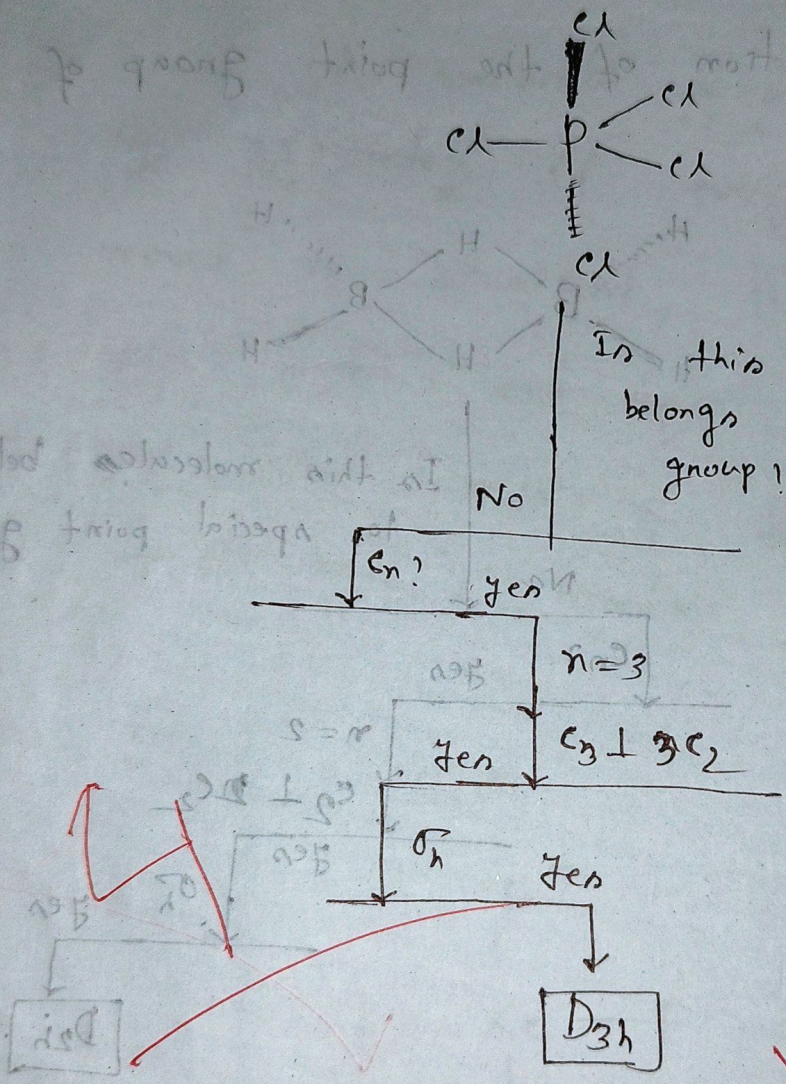
σ_h

yes

D_{2h}

Therefore, B_2H_6 belongs to D_{2h} point group.

Determination of the point group of PCl_5,



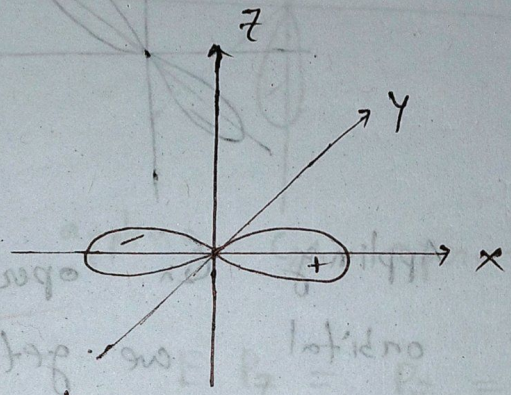
Therefone, PCl_5 belongs to D_{3h} point group.

Ans to the Q.No.c

Let, C_2 axis as the z axis,

σ_h to be xy plane

For P_x onbital:



After Appling the C_{2h} operation then we get,

$$E \cdot P_x = P_x = 1$$

$$C_2 \cdot P_x = -P_x = -1$$

$$i P_x = -P_x = -1$$

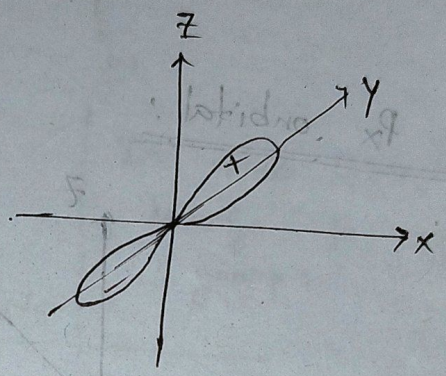
$$\sigma_h P_x = P_x = 1$$

Therefore the P_x onbital nepresent as,

C_{2h}	E	C_2	i	σ_h	
P_x B_u	1	−1	−1	1	x

Therefore, P_x - orbital belongs to B_u representation.

For, P_y - orbital:



After Appling C_{2h} operation to the P_y orbital we get,

$$E\,P_y = P_y = 1$$

$$C_2\,P_y = -P_y = -1$$

$$i\,P_y = -P_y = -1$$

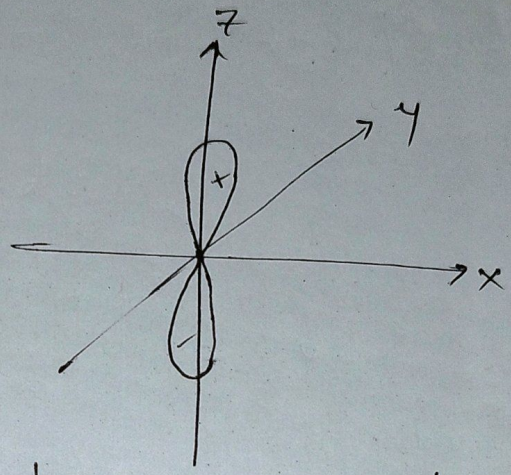
$$\sigma_h\,P_y = P_y = 1$$

Therefore the P_y orbital represent as,

C_{2h}	E	C_2	i	σ_h
P_y B_u	1	-1	-1	1

P_y onbital Belongs to Bu nepnesentation.

For, P_z -onbital:



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After appling C_{2h} operation,

$$E\ P_z = P_z = 1$$

$$C_2\ P_z = P_z = 1$$

$$i\ P_z = -P_z = -1$$

$$\sigma_h\ P_z = -P_z = -1$$

The P_z onbital nepnesent as,

C_{2h}	E	C_2	i	σ_h
P_z Au	1	1	-	-1

Thenefone, P_z onbital belongs to Au nepnesentation.